

OPERATIONAL INTELLIGENCE REPORT

NIGERIA

POST-HARVEST AGRICULTURAL INTELLIGENCE DASHBOARD

Reducing Spoilage · Protecting Revenue · Empowering Youth Entrepreneurs



PREPARED BY

IROAGBA VALENTINE CHIBUEZE

Agricultural Data Intelligence · Nigeria · 2023–2024 Dataset

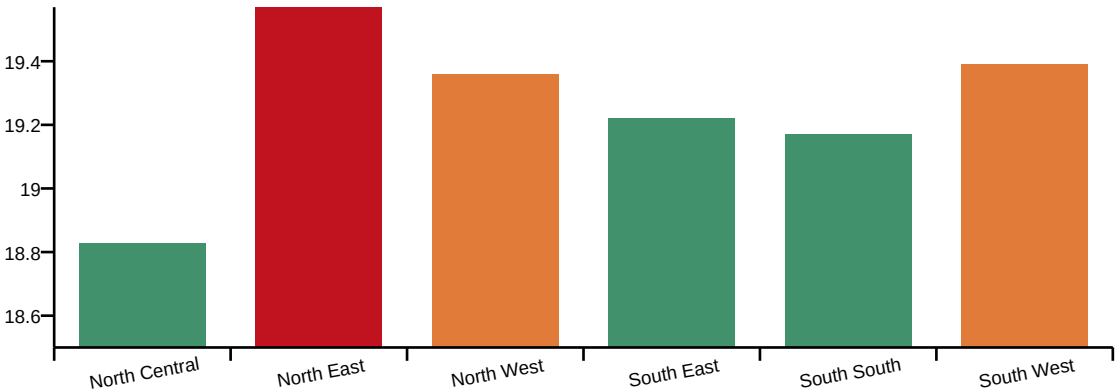
SECTION 1 — EXECUTIVE OVERVIEW

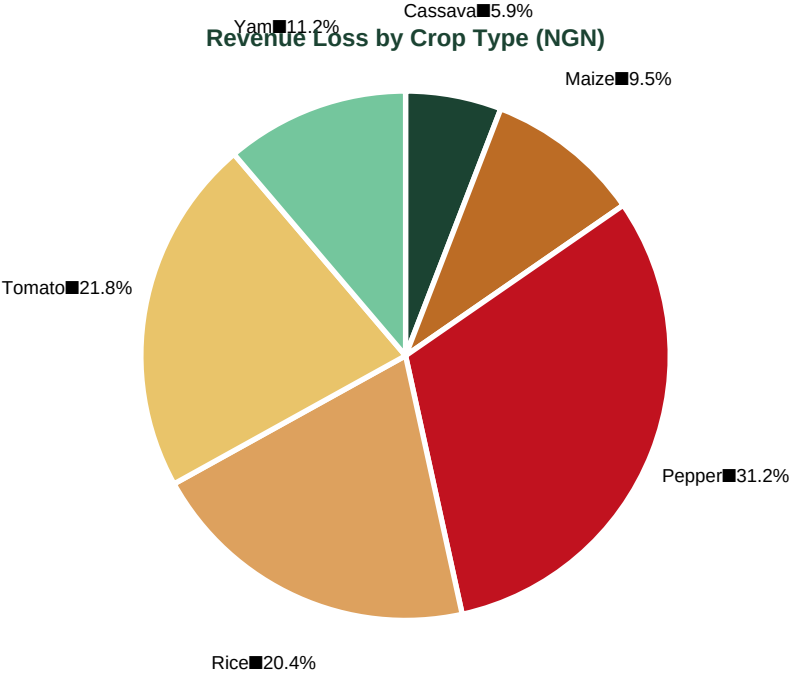
Key Performance Indicators · Regional & Crop Analysis

CRITICAL ALERT: Nigeria's post-harvest loss rate is 19.25% — nearly double the FAO benchmark of 10%. Across 5,000 farming operations, total revenue loss stands at █41.4 million NGN. The North East zone carries the highest risk. Only 39.7% of farmers use post-harvest technology and just 28.9% have received formal training.

KPI	Value	Benchmark	Status
Total Harvest Volume	1,004,961 kg	—	BASELINE
Total Spoilage	193,071 kg	Minimize	CRITICAL
Average PHL Rate	19.25%	< 10%	CRITICAL
Total Revenue Loss	█41,415,461	Minimize	CRITICAL
Avg Rev Loss / Farmer	■8,283	Minimize	WARNING
Technology Adoption	39.7%	> 70%	WARNING
Training Coverage	28.9%	> 60%	CRITICAL
Youth Participation	39.9%	> 50%	WARNING
Avg Storage Duration	10.0 days	< 7 days	WARNING
Avg Temperature	29.9°C	< 25°C	CRITICAL

Average PHL Rate by Region (%)





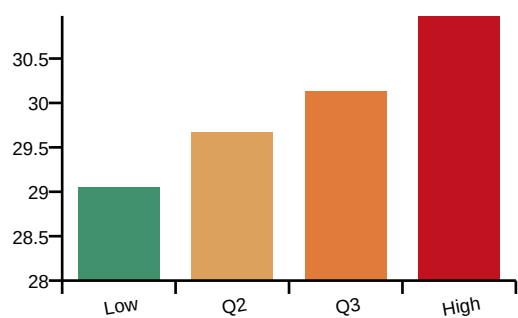
SECTION 2 — SPOILAGE & ROOT-CAUSE ANALYSIS

Environmental Drivers · Heatmap · Seasonal

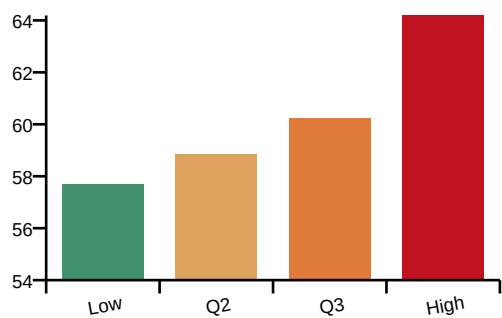
Root Cause: Farms operate at average temperatures of 29.9°C (optimal: <25°C) and humidity of 60.2% (optimal: <55%). High-spoilage operations show nearly 2°C higher temperature and 6.5% more humidity than low-spoilage operations — compounding decay at every stage.

Environmental Conditions by Spoilage Severity

Avg Temperature (°C) by Spoilage Quartile



Avg Humidity (%) by Spoilage Quartile



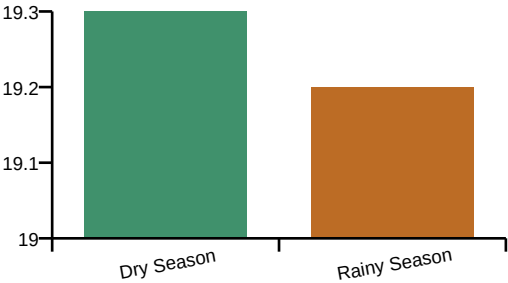
Region × Crop PHL Heatmap (%)

Darker green = higher spoilage risk. Identify zone-crop combinations requiring priority intervention.

Region	Cassava	Maize	Pepper	Rice	Tomato	Yam
North Central	19.62%	18.81%	18.83%	19.20%	18.61%	18.50%
North East	19.28%	19.14%	19.95%	19.55%	19.84%	19.65%
North West	19.59%	19.19%	18.89%	19.68%	19.45%	19.39%
South East	18.98%	19.15%	18.27%	19.60%	18.59%	20.38%
South South	20.01%	19.29%	19.76%	19.35%	18.78%	

South West	19.27%	18.98%	19.78%	20.16%	19.54%	18.58%
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Avg PHL % by Season

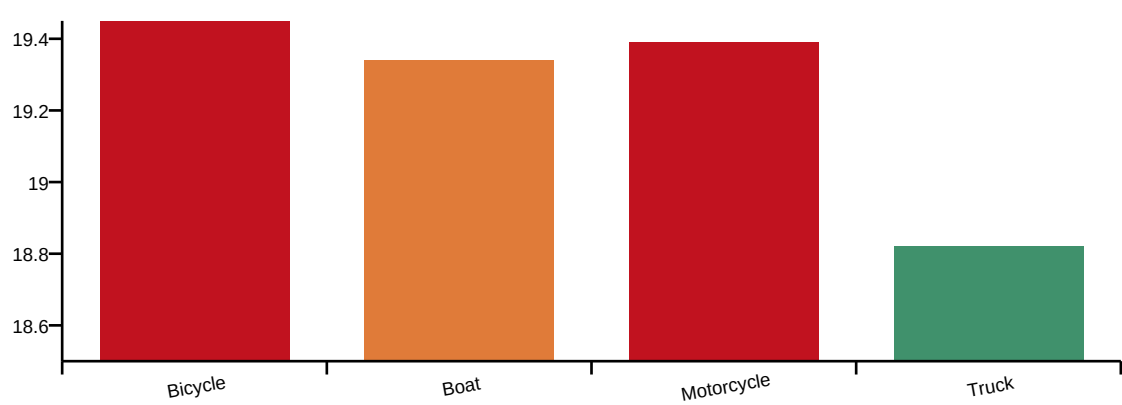


SECTION 3 — LOGISTICS & STORAGE

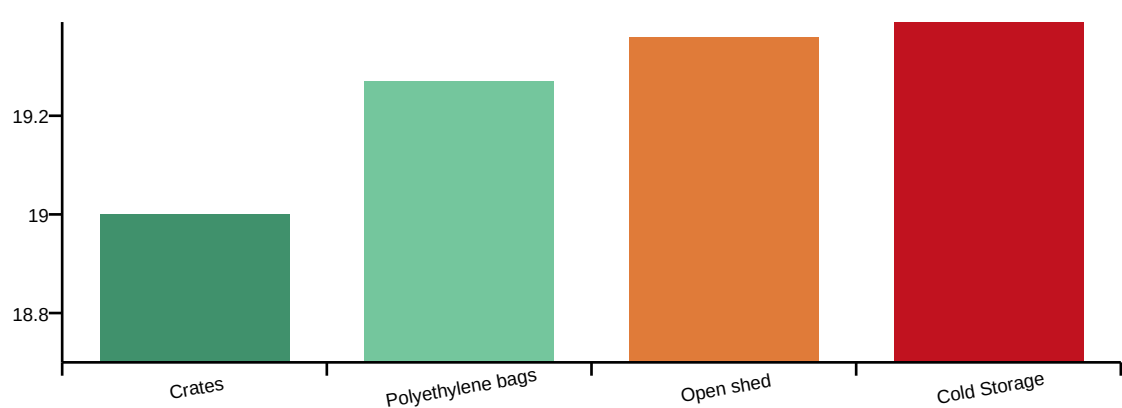
Transport Mode · Storage Method · Market Access

Key Finding: Truck transport delivers the lowest PHL (18.82%) and lowest revenue loss (■7,917 avg). Motorcycle transport causes the highest financial loss (■8,711 avg). Cold Storage has the WORST PHL (19.39%) — a signal of infrastructure failure, not conceptual failure.

Avg PHL % by Transport Mode



Avg PHL % by Storage Method



Transport Type	Avg PHL %	Avg Rev Loss (■)	Avg Distance	Performance
Bicycle	19.45%	■8,254	19.7 km	HIGH RISK
Boat	19.34%	■8,264	19.9 km	MODERATE
Motorcycle	19.39%	■8,711	20.7 km	HIGH RISK
Truck	18.82%	■7,917	19.4 km	BEST

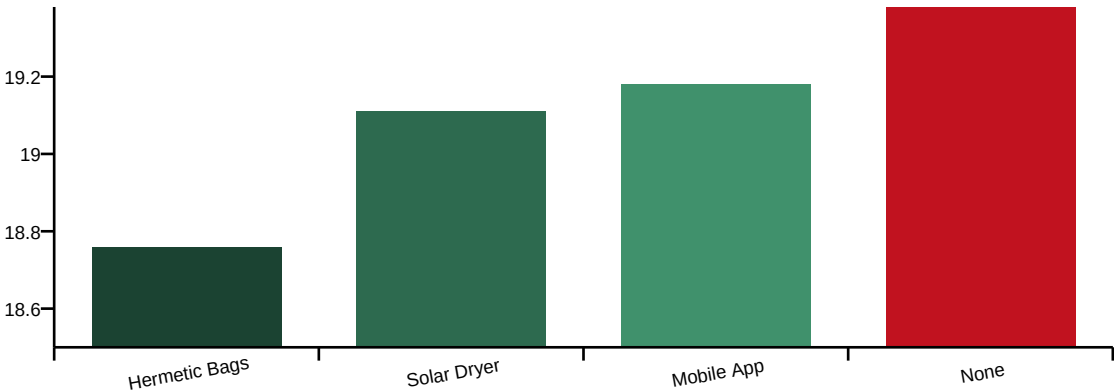
SECTION 4 — INTERVENTIONS

Technology · Training · Youth

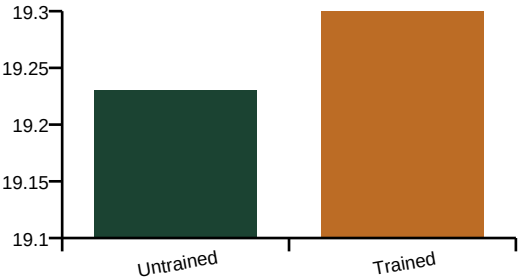
Hermetic Bags are the most impactful technology (PHL: 18.76%). Mobile Apps reduce financial losses most (₦7,940 avg). Training shows negligible direct PHL impact (+0.07pp), suggesting quality issues. Youth underperformance reflects structural constraints, not incompetence.

Metric	Current	Target	Gap
Technology Adoption	39.7%	> 70%	30.3% gap
Training Coverage	28.9%	> 60%	31.1% gap
Youth Participation	39.9%	> 50%	10.1% gap

Avg PHL % by Technology Used



Avg PHL %: Trained vs Untrained Farmers



SECTION 5 — SIMPLIFIED INSIGHTS & STRATEGIC RECOMMENDATIONS

Plain Language Summary · 8 Evidence-Based Interventions

If the top 4 interventions are deployed within 12 months, a projected 4–6 percentage point PHL reduction is achievable — recovering █8–12 million NGN in protected farmer revenue annually.

Key Insights — In Plain Language

1-in-5 kg of food is lost	Every 5 bags harvested, nearly 1 bag is lost before reaching market. This is preventable.
It's too hot and humid	Produce stored at 30°C+ should be below 25°C. Heat is rotting food faster than farmers can sell it.
Cold storage isn't working	Cold storage has the WORST spoilage rate despite being most expensive — equipment isn't maintained.
Motorbikes are expensive	Motorcycle transport causes the highest financial loss (█8,711 avg). Cooperative trucks save █794/farmer.
Pepper costs the most	█12.9M lost — because pepper is expensive and highly perishable without proper cold chain.
Hermetic bags + mobile apps win	Hermetic bags reduce PHL to 18.76%. Mobile apps reduce financial losses most. Best ROI available.
Youth need a full support package	Youth operations underperform not from lack of skill, but from lack of tools, training & market access.

Strategic Recommendations — 8 Interventions

01	URGENT Scale Hermetic Bag Distribution Subsidize hermetic bags in North East, South West & North West zones. Creates airtight environments that stop insects and decay. Best-performing tech in the dataset. <i>Impact: -0.62pp PHL · Targets 3,017 unequipped farmers</i>
02	URGENT Emergency Training Scale-Up Deploy structured, outcome-linked training in the 3 highest-PHL zones. Bundle with equipment. Target: 60%+ farmer coverage within 90 days. <i>Impact: Training 28.9% → 60%+ · Behaviour change within 1 season</i>
03	HIGH Cold Storage Infrastructure Audit Audit all cold storage for power reliability, thermostat accuracy, load management. Cold storage currently has the WORST PHL (19.39%) — fix before more investment. <i>Impact: Could recover Cold Storage from worst to best-in-class performance</i>

04

HIGH

Motorcycle → Truck Cooperative

Create community aggregation hubs every 10–15 km for truck collection. Cooperative trucking reduces cost and spoilage simultaneously.

Impact: PHL: 19.39% → 18.82% · ■794 avg recovery per farmer

05

MEDIUM

Mobile Agricultural Platform

Deploy platform with market prices, weather alerts, logistics booking & advisory. Mobile app users achieve lowest revenue loss (■7,940 avg).

Impact: Highest scalability — near-zero marginal cost per additional farmer

06

MEDIUM

Solar Dryer Expansion

Expand solar dryer access for Tomato, Pepper & Cassava farmers. Enables value-added drying, longer shelf life, and higher prices.

Impact: Addresses spoilage reduction and value creation for high-loss crops

07

LONG-TERM

Youth Enterprise Programme

Dedicated support ecosystem: subsidized equipment, certified training, market guarantees, cooperative buying, crop insurance for 39.9% youth-involved operations.

Impact: Transforms the largest at-risk group into a productivity engine

08

LONG-TERM

Regional Intelligence Centres

Permanent centres in North East, South West & North West — hubs for training delivery, tech distribution, cold chain oversight, and data collection.

Impact: Institutional capacity for sustained, data-driven improvement

SECTION 6 — FINAL STRATEGIC CONCLUSION

Summary for Decision-Makers

Nigeria's post-harvest loss challenge is solvable within a 3–5 year horizon with coordinated intervention across technology access, training quality, logistics infrastructure, and market linkage.

The 19.25% average PHL rate affecting 5,000 farming operations and generating ■41.4 million in documented losses represents household income erosion, food system inefficiency, and a structural barrier to agricultural economic development. None of these are impossible problems — they are organizational, investment, and policy challenges with clear evidence-based solutions. The strategic will to act is the only remaining variable.

Deploy hermetic bags and scale training within 90 days. Audit cold storage and launch transport cooperatives within 12 months. Build the youth enterprise ecosystem and regional intelligence centres over 24–36 months. The return on each intervention is documented, measurable, and operationally credible.

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